

-- 1. Trend prodaje po proizvodima

-- 1. Sales trend by products

-- Pronađi top 5 proizvoda po rastu ukupnog prihoda mesec po mesec u poslednjih 12 meseci koristeći window funkcije.

-- Find the top 5 products by month-over-month revenue growth in the last 12 months using window functions.

WITH monthly_revenue_last_12 AS (

SELECT

p.productName,

YEAR(o.orderDate) AS Year_,

MONTH(o.orderDate) AS Month_,

SUM(od.quantity * od.unitPrice) AS TotalRevenue

FROM orders o

LEFT JOIN order_details od ON o.orderID = od.orderID

LEFT JOIN products p ON od.productID = p.productID

WHERE o.orderDate >= '2014-06-01' AND o.orderDate <= '2015-05-31'

GROUP BY p.productName, YEAR(o.orderDate), MONTH(o.orderDate)

),

products_revenue_growth AS (

SELECT

mr.productName,

Year_,

Month_,

(TotalRevenue - LAG(TotalRevenue) OVER(PARTITION BY
mr.productName ORDER BY Year_,Month_)) AS RevenueGrowth

FROM

monthly_revenue_last_12 mr

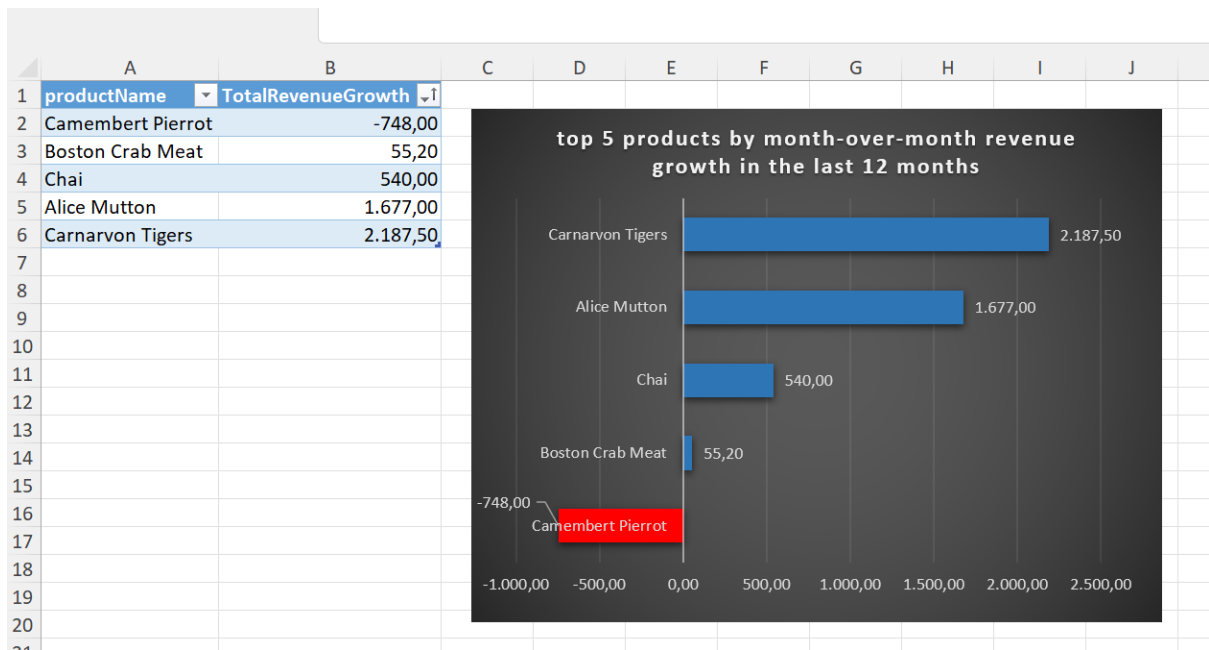
ORDER BY Year_, Month_

```

)
SELECT
    prg.productName,
    SUM(IFNULL(RevenueGrowth,0)) AS TotalRevenueGrowth
FROM
    products_revenue_growth prg
GROUP BY
    prg.productName
LIMIT 5;

```

Result Grid		Filter Rows:
	productName	TotalRevenueGrowth
▶	Alice Mutton	1677
	Boston Crab Meat	55.200000000000027
	Camembert Pierrot	-748
	Carnarvon Tigers	2187.5
	Chai	540



-- 2. Mesečna prodaja po zaposlenima

-- 2. Monthly sales by employees

-- Izračunaj mesečni rast prodaje po zaposlenima koristeći LAG funkciju.

-- Calculate monthly sales growth by employees using the LAG function.

WITH monthly_revenue_by_employees AS (

SELECT

 e.employeeName,

 YEAR(o.orderDate) AS Year_,

 MONTH(o.orderDate) AS Month_,

 SUM(od.quantity * od.unitPrice) AS TotalRevenue

FROM orders o

LEFT JOIN order_details od ON o.orderID = od.orderID

LEFT JOIN employees e ON o.employeeID = e.employeeID

GROUP BY e.employeeName, YEAR(o.orderDate), MONTH(o.orderDate)

)

SELECT

 mre.employeeName,

 Year_,

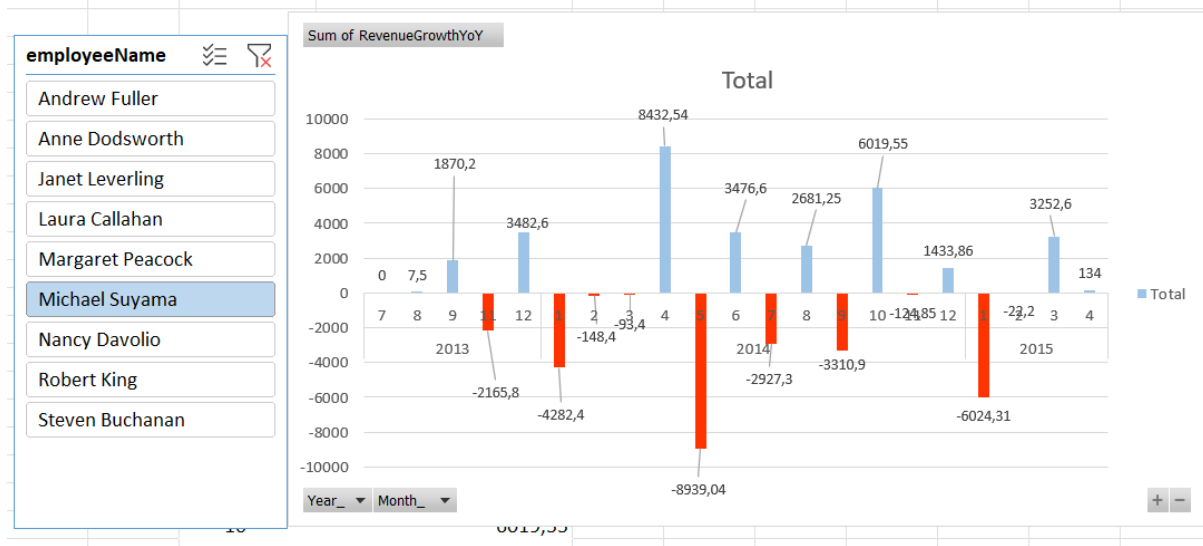
 Month_,

 TotalRevenue - LAG(TotalRevenue) OVER (PARTITION BY mre.employeeName ORDER
BY Year_,Month_) AS RevenueGrowthYoY

FROM

 monthly_revenue_by_employees mre;

Result Grid				
	Filter Rows:		Export:	
	employeeName	Year_	Month_	RevenueGrowthYoY
▶	Andrew Fuller	2013	7	NULL
	Andrew Fuller	2013	8	638
	Andrew Fuller	2013	9	1136.7999999999997
	Andrew Fuller	2013	10	2774.9
	Andrew Fuller	2013	11	-966.6999999999998
	Andrew Fuller	2013	12	1658.1000000000001



-- 3. Top kupci po prihodu

-- 3. Top customers by revenue

-- Pronađi top 10 kupaca po ukupnom prihodu u poslednjih 6 meseci koristeći CTE i ROW_NUMBER().

-- Find the top 10 customers by total revenue in the last 6 months using CTE and ROW_NUMBER().

WITH total_revenue_by_customers AS (

SELECT

c.companyName,

SUM(od.quantity * od.unitPrice) AS TotalRevenue

FROM orders o

LEFT JOIN order_details od ON o.orderID = od.orderID

LEFT JOIN customers c ON o.customerID = c.customerID

GROUP BY c.companyName

)

SELECT

trc.companyName,

trc.TotalRevenue,

ROW_NUMBER() OVER (ORDER BY trc.TotalRevenue DESC) AS ROWNUMBER

FROM

total_revenue_by_customers trc

ORDER BY

ROWNUMBER

LIMIT

10;

Result Grid	Filter Rows:	Export:	Wrap Cell Cor
companyName	TotalRevenue	ROWNUMBER	
QUICK-Stop	117483.39000000001	1	
Save-a-lot Markets	115673.39	2	
Ernst Handel	113236.68	3	
Hungry Owl All-Night Grocers	57317.39	4	
Rattlesnake Canyon Grocery	52245.9	5	
Hanari Carnes	34101.15	6	
Folk och fä HB	32555.55	7	
Mère Paillard	32203.90	8	
Königlich Essen	31745.75	9	
Queen Cozinha	30226.10	10	

	A	B	C	D	E	F
1	companyName	COMPANY NAME AND RANK	TotalRevenue	ROWNUMBER		
2	Queen Cozinha	10. Queen Cozinha	30.226,10	10		
3	Königlich Essen	9. Königlich Essen	31.745,75	9		
4	Mère Paillard	8. Mère Paillard	32.203,90	8		
5	Folk och fä HB	7. Folk och fä HB	32.555,55	7		
6	Hanari Carnes	6. Hanari Carnes	34.101,15	6		
7	Rattlesnake Canyon Grocery	5. Rattlesnake Canyon Grocery	52.245,90	5		
8	Hungry Owl All-Night Grocers	4. Hungry Owl All-Night Grocers	57.317,39	4		
9	Ernst Handel	3. Ernst Handel	113.236,68	3		
10	Save-a-lot Markets	2. Save-a-lot Markets	115.673,39	2		
11	QUICK-Stop	1. QUICK-Stop	117.483,39	1		
12						
13						
14						
15						
16						
17						
18						
19						
20						
21						
22						
23						
24						
25						
26						

top 10 customers by total revenue in the last 6 months

1. QUICK-Stop	117.483,39
2. Save-a-lot Markets	115.673,39
3. Ernst Handel	113.236,68
4. Hungry Owl All-Night Grocers	57.317,39
5. Rattlesnake Canyon Grocery	52.245,90
6. Hanari Carnes	34.101,15
7. Folk och fä HB	32.555,55
8. Mère Paillard	32.203,90
9. Königlich Essen	31.745,75
10. Queen Cozinha	30.226,10

-- 4. Najprofitabilniji šiper

-- 4. Most profitable shipper

-- Odredi šipera sa najvećim neto prihodom po zemlji koristeći SUM() CASE(WHEN) - pivot.

-- Identify the shipper with the highest net revenue per country using SUM() CASE(WHEN) - pivot.

SELECT

 DISTINCT(c.country)

FROM

 customers c;

SELECT

 s.companyName,

 SUM(CASE WHEN c.country = 'Germany' THEN od.quantity * od.unitPrice ELSE 0
END) AS TotalRevenueGermany,

 SUM(CASE WHEN c.country = 'Mexico' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenueMexico,

 SUM(CASE WHEN c.country = 'UK' THEN od.quantity * od.unitPrice ELSE 0 END) AS
TotalRevenueUK,

 SUM(CASE WHEN c.country = 'Sweden' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenueSweden,

 SUM(CASE WHEN c.country = 'France' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenueFrance,

 SUM(CASE WHEN c.country= 'Spain' THEN od.quantity * od.unitPrice ELSE 0 END) AS
TotalRevenueSpain,

 SUM(CASE WHEN c.country = 'Canada' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenueCanada,

 SUM(CASE WHEN c.country = 'Argentina' THEN od.quantity * od.unitPrice ELSE 0
END) AS TotalRevenueArgentina,

```

SUM(CASE WHEN c.country = 'Switzerland' THEN od.quantity * od.unitPrice ELSE 0
END) AS TotalRevenueSwitzerland,

SUM(CASE WHEN c.country = 'Brazil' THEN od.quantity * od.unitPrice ELSE 0 END) AS
TotalRevenueBrazil,

SUM(CASE WHEN c.country = 'Austria' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenueAustria,

SUM(CASE WHEN c.country = 'Italy' THEN od.quantity * od.unitPrice ELSE 0 END) AS
TotalRevenueItaly,

SUM(CASE WHEN c.country = 'Portugal' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenuePortugal,

SUM(CASE WHEN c.country = 'USA' THEN od.quantity * od.unitPrice ELSE 0 END) AS
TotalRevenueUSA,

SUM(CASE WHEN c.country = 'Venezuela' THEN od.quantity * od.unitPrice ELSE 0
END) AS TotalRevenueVenezuela,

SUM(CASE WHEN c.country = 'Ireland' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenueIreland,

SUM(CASE WHEN c.country = 'Belgium' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenueBelgium,

SUM(CASE WHEN c.country = 'Norway' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenueNorway,

SUM(CASE WHEN c.country = 'Denmark' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenueDenmark,

SUM(CASE WHEN c.country = 'Finland' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenueFinland,

SUM(CASE WHEN c.country = 'Poland' THEN od.quantity * od.unitPrice ELSE 0 END)
AS TotalRevenuePoland

FROM orders o

LEFT JOIN
    order_details od ON o.orderID = od.orderID

LEFT JOIN
    customers c ON o.customerID = c.customerID

LEFT JOIN

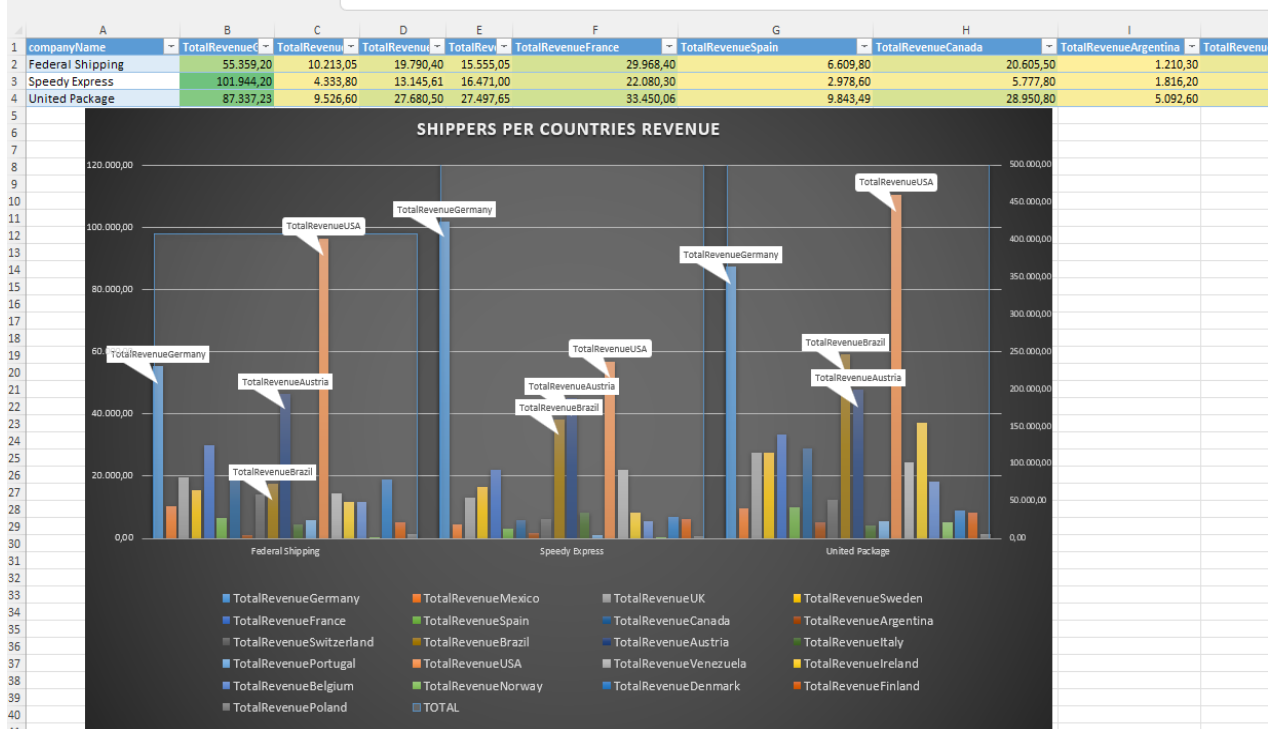
```

shippers s ON o.shipperID = s.shipperID

GROUP BY

s.companyName;

Result Grid						
Filter Rows:		Export:		Wrap Cell Content:		
companyName	TotalRevenueGermany	TotalRevenueMexico	TotalRevenueUK	TotalRevenueSweden	TotalRevenueFrance	Total
Federal Shipping	55359.2	10213.05	19790.4	15555.05	29968.4	6609.
Speedy Express	101944.20000000001	4333.8	13145.609999999999	16471	22080.3	2978.
United Package	87337.23000000001	9526.599999999999	27680.5	27497.649999999998	33450.06	9843.



-- 5. Analiza prosečne količine po zaposlenom

-- 5. Average order quantity per employee analysis

-- Za svakog zaposlenog izračunaj prosečnu količinu narudžbi po mesecu i rangiraj ih po mesecu.

-- Calculate the average order quantity per employee per month and rank them by month.

SELECT

e.employeeName,

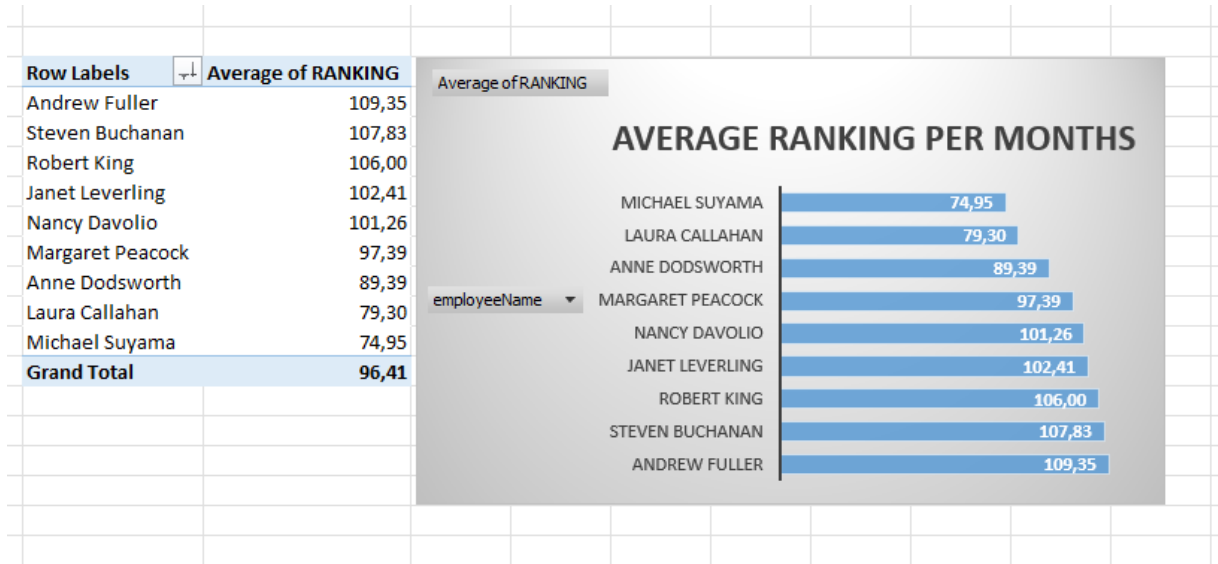
YEAR(o.orderDate) AS Year_,

```

MONTH(o.orderDate) AS Month_,
AVG(od.quantity) AS AvgQuantitySold,
RANK() OVER(ORDER BY AVG(od.quantity)) AS RANKING
FROM orders o
LEFT JOIN order_details od ON o.orderID = od.orderID
LEFT JOIN employees e ON o.employeeID = e.employeeID
GROUP BY e.employeeName, YEAR(o.orderDate), MONTH(o.orderDate)
ORDER BY RANKING ASC;

```

Result Grid					
Filter Rows:					
Export: Export Wrap					
	employeeName	Year_	Month_	AvgQuantitySold	RANKING
▶	Anne Dodsworth	2014	5	4.0000	1
	Anne Dodsworth	2014	7	4.0000	1
	Andrew Fuller	2014	8	5.0000	3
	Laura Callahan	2013	11	6.5000	4
	Laura Callahan	2015	2	6.5000	4
	Laura Callahan	2013	7	6.5000	6



-- 6. Kumulativni prihod po proizvodu

-- 6. Cumulative revenue by product

-- Prikaži kumulativni prihod po proizvodu kroz vreme koristeći SUM() OVER (ORDER BY ...).

-- Show cumulative revenue by product over time using SUM() OVER (ORDER BY ...).

WITH monthly_product_revenue AS (

SELECT

p.productName,

YEAR(o.orderDate) AS Year_,

MONTH(o.orderDate) AS Month_,

SUM(od.quantity * od.unitPrice) AS MonthlyRevenue

FROM orders o

LEFT JOIN order_details od ON o.orderID = od.orderID

LEFT JOIN products p ON od.productID = p.productID

GROUP BY

p.productName,

YEAR(o.orderDate),

MONTH(o.orderDate)

)

SELECT

productName,

Year_,

Month_,

MonthlyRevenue,

SUM(MonthlyRevenue) OVER (

PARTITION BY productName

ORDER BY Year_, Month_

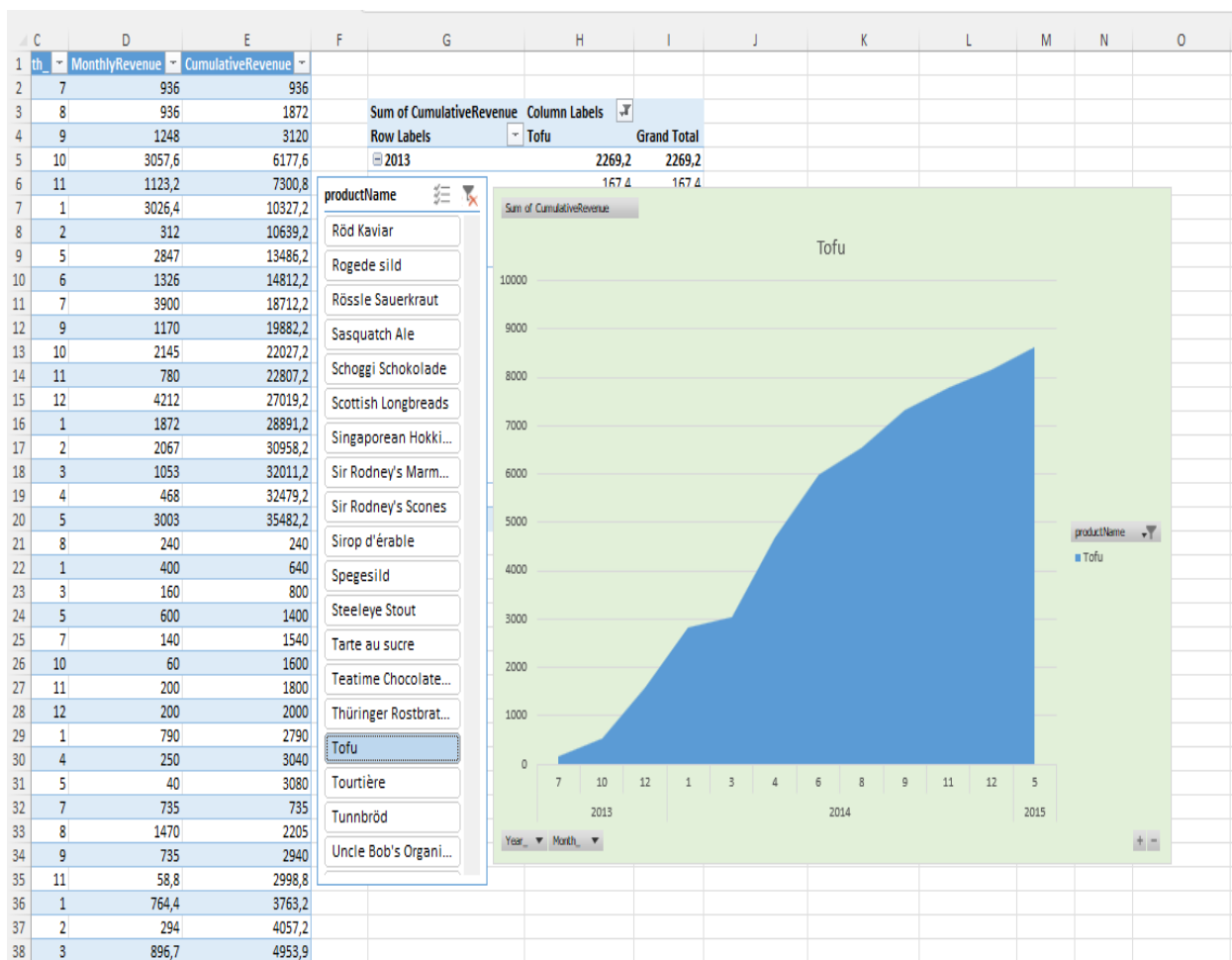
) AS CumulativeRevenue

FROM monthly_product_revenue

ORDER BY productName, Year_, Month_;

Result Grid					
Filter Rows:		Export:  Wrap Cell Content: 			
	productName	Year_	Month_	MonthlyRevenue	CumulativeRevenue
▶	Alice Mutton	2013	7	936	936
	Alice Mutton	2013	8	936	1872
	Alice Mutton	2013	9	1248	3120
	Alice Mutton	2013	10	3057.6	6177.6
	Alice Mutton	2013	11	1123.2	7300.8
	Alice Mutton	2014	1	3026.4	10327.2

Result 41 x



-- 7. Najveći skok prihoda po kategoriji

-- 7. Highest revenue jump by category

-- Pronađi kategoriju proizvoda koja je imala najveći rast / najmanji pad prihoda poslednjeg meseca u odnosu na prethodni mesec koristeći CTE i LAG().

-- Find the product category with the highest revenue growth / the lowest revenue fall of last month compared to the previous month using CTE and LAG().

WITH monthly_category_revenue AS (

SELECT

c.categoryName,

YEAR(o.orderDate) AS Year_,

MONTH(o.orderDate) AS Month_,

SUM(od.quantity * od.unitPrice) AS MonthlyRevenue

FROM orders o

LEFT JOIN order_details od ON o.orderID = od.orderID

LEFT JOIN products p ON od.productID = p.productID

LEFT JOIN categories c ON p.categoryID = c.categoryID

GROUP BY

c.categoryName,

YEAR(o.orderDate),

MONTH(o.orderDate)

),

revenue_growth AS (

SELECT

categoryName,

Year_,

Month_,

MonthlyRevenue,

```
MonthlyRevenue - LAG(MonthlyRevenue) OVER (PARTITION BY categoryName
ORDER BY Year_, Month_) AS MonthlyRevenueGrowth
```

```
FROM monthly_category_revenue
```

```
)
```

```
SELECT *
```

```
FROM revenue_growth
```

```
WHERE (Year_, Month_) = (
```

```
SELECT Year_, Month_
```

```
FROM revenue_growth
```

```
ORDER BY Year_ DESC, Month_ DESC
```

```
LIMIT 1
```

```
)
```

```
ORDER BY MonthlyRevenue DESC
```

```
LIMIT 1;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	categoryName	Year_	Month_	MonthlyRevenue	MonthlyRevenueGrowth
▶	Beverages	2015	5	4597	-19873

-- 8. Prosečan prihod po kupcu i mesecu

-- 8. Average revenue per customer per month

-- Izračunaj prosečan prihod po kupcu po mesecu i rangiraj kupce unutar svake zemlje po proseku prihoda koristeći RANK().

-- Calculate average revenue per customer per month and rank customers within each country using RANK().

```
WITH customer_monthly_revenue AS (
```

```
SELECT
```

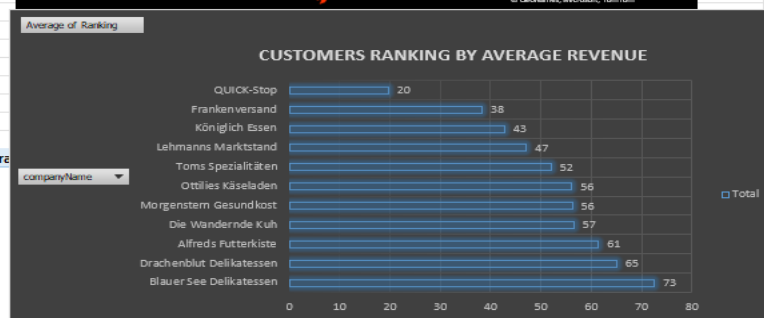
```
c.country,  
c.companyName,  
YEAR(o.orderDate) AS Year_,  
MONTH(o.orderDate) AS Month_,  
AVG(od.quantity * od.unitPrice) AS AvgRevenue  
FROM orders o  
LEFT JOIN order_details od ON o.orderID = od.orderID  
LEFT JOIN customers c ON o.customerID = c.customerID  
GROUP BY  
c.country,  
c.companyName,  
YEAR(o.orderDate),  
MONTH(o.orderDate)  
)
```

```
SELECT  
country,  
companyName,  
Year_,  
Month_,  
AvgRevenue,  
RANK() OVER (PARTITION BY country ORDER BY AvgRevenue DESC) AS Ranking  
FROM customer_monthly_revenue  
ORDER BY  
country,  
Ranking;
```

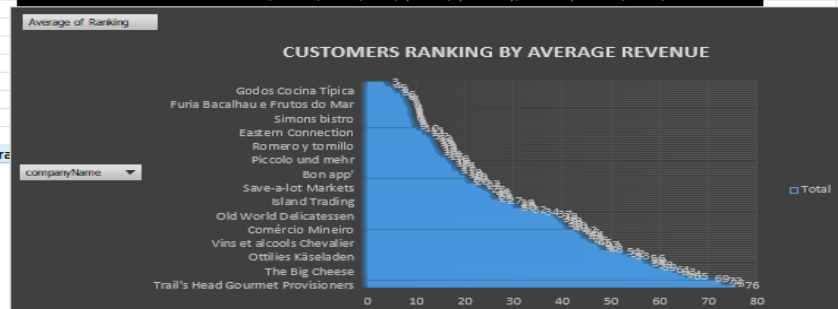
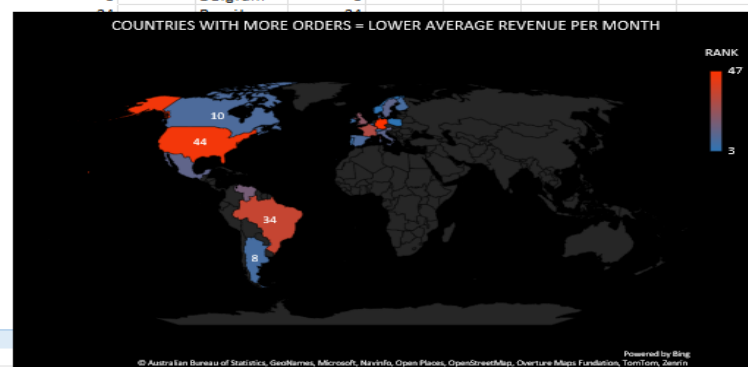
Result Grid						
Filter Rows:		Export:		Wrap Cell Content:		
	country	companyName	Year_	Month_	AvgRevenue	Ranking
▶	Argentina	Rancho grande	2015	1	466	1
	Argentina	Océano Atlántico Ltda.	2015	3	428.7142857142857	2
	Argentina	Cactus Comidas para llevar	2015	3	322.4	3
	Argentina	Cactus Comidas para llevar	2015	1	238.5	4
	Argentina	Rancho grande	2014	10	235.33333333333334	5
	Argentina	Rancho grande	2015	3	238.5	6

Result 43 x

country
Argentina
Austria
Belgium
Brazil
Canada
Denmark
Finland
France
Germany
Ireland
Italy
Mexico
Norway
Poland
Portugal
Spain
Sweden
Switzerland
UK
USA
Venezuela



country
Argentina
Austria
Belgium
Brazil
Canada
Denmark
Finland
France
Germany
Ireland
Italy
Mexico
Norway
Poland
Portugal
Spain
Sweden
Switzerland
UK
USA
Venezuela



-- 9. Efikasnost prodaje po jedinici tereta

-- 9. Sales and Freight Efficiency

-- Meseci kada je prihod po jedinici tovara bio najveći u svakoj zemlji

-- Months when revenue per freight unit was the highest for each country.

WITH order_totals AS (

SELECT

o.orderID,

c.country,

YEAR(o.orderDate) AS Year_,

MONTH(o.orderDate) AS Month_,

SUM(od.quantity * od.unitPrice) AS Revenue,

o.freight AS Freight

FROM orders o

LEFT JOIN order_details od ON o.orderID = od.orderID

LEFT JOIN customers c ON o.customerID = c.customerID

GROUP BY

o.orderID,

c.country,

YEAR(o.orderDate),

MONTH(o.orderDate),

o.freight -- we must have one freight from orders parallel to SUM(od.quantity *
od.unitPrice) from order_details (because of that it is in group by...)

),

monthly_revenue_per_freight AS (

SELECT

country,

Year_,

Month_

ROUND(SUM(Revenue) / SUM(Freight), 2) AS RevenuePerFreightUnit -- when we have one freight row to one Revenue row, it`s time for aggregation

FROM order_totals

GROUP BY

country,

Year_

Month_

)

SELECT

country,

Year_

Month_

RevenuePerFreightUnit,

RANK() OVER (PARTITION BY country ORDER BY RevenuePerFreightUnit DESC) AS RankingRevPerFreUnit

FROM monthly_revenue_per_freight

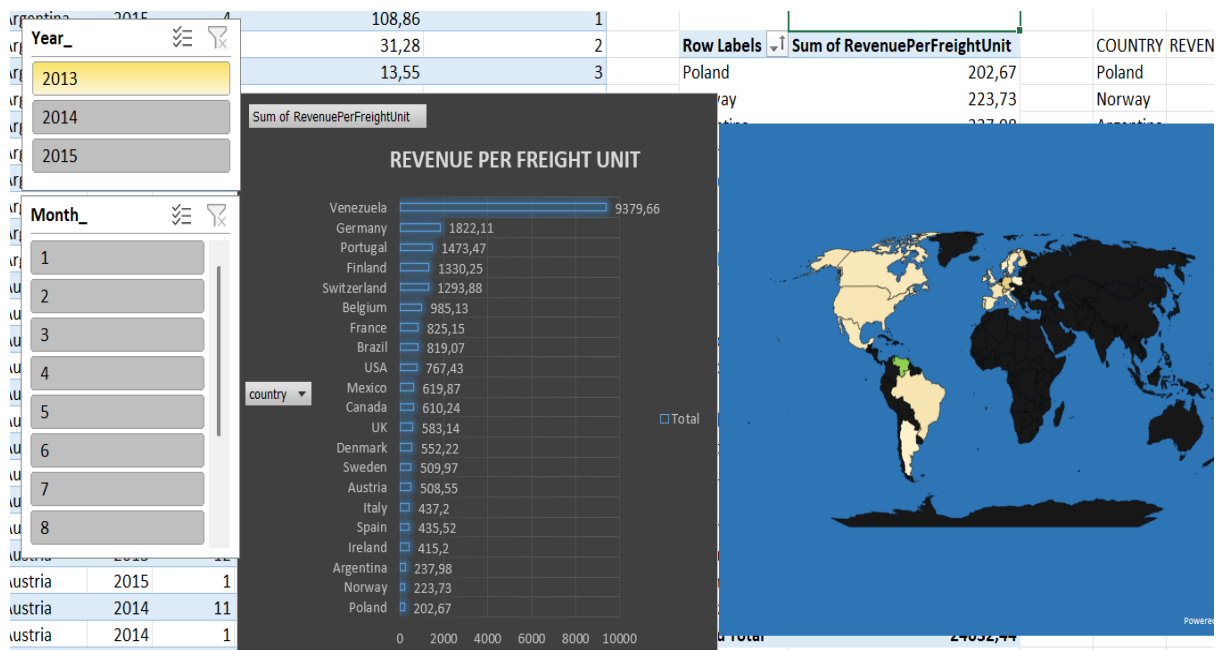
ORDER BY

country,

RankingRevPerFreUnit;

	country	Year_	Month_	RevenuePerFreightUnit	RankingRevPerFreUnit
▶	Argentina	2015	4	108.86	1
	Argentina	2014	10	31.28	2
	Argentina	2014	5	13.55	3
	Argentina	2014	4	13.1	4
	Argentina	2015	2	12.77	5
	Argentina	2015	1	12.74	6

Result 44 x



-- 10. Najlošije prodavani proizvodi

-- 10. Worst-selling products

-- Prikaži top 5 proizvoda sa najmanjom količinom prodatih jedinica poslednjih 6 meseci koristeći CTE.

-- Show the top 5 products with the lowest sold quantity in the last 6 months using CTE.

WITH monthly_product_revenue AS (

SELECT

p.productName,

SUM(od.quantity) AS TotalQuantity

FROM orders o

LEFT JOIN order_details od ON o.orderID = od.orderID

LEFT JOIN products p ON od.productID = p.productID

WHERE o.orderDate >= '2014-12-1' and o.orderDate <= '2015-5-31'

GROUP BY

p.productName

)

SELECT

```

*
FROM
    monthly_product_revenue mpr
ORDER BY
    mpr.TotalQuantity ASC
LIMIT
    5;

```

Result Grid		Filter Rows:	Export
	productName	TotalQuantity	
▶	Mishi Kobe Niku	3	
	Chocolate	8	
	Röd Kaviar	12	
	Tofu	36	
	Louisiana Hot Spiced Okra	51	

Result 45 ▼

-- 11. Sezonska analiza prodaje

-- 11. Seasonal sales analysis

-- Izračunaj ukupan prihod po kvartalu i identifikuj sezonske trendove pivotirajući po QUARTERima.

-- Calculate total revenue per quarter and identify seasonal trends pivoting quarters revenue per years.

```

WITH revenue_by_quarter_and_month AS (
    SELECT
        YEAR(o.orderDate) AS Year_,
        QUARTER(o.orderDate) AS Quarter_,
        ROUND(SUM(od.quantity * od.unitPrice),2) AS QuarterRevenue
    FROM orders o
    LEFT JOIN order_details od ON o.orderID = od.orderID
    GROUP BY
        YEAR(o.orderDate),
        QUARTER(o.orderDate)

```

)

SELECT

rqm.Year_,

SUM(CASE WHEN Quarter_ = 1 THEN QuarterRevenue ELSE 0 END) AS Q1,

SUM(CASE WHEN Quarter_ = 2 THEN QuarterRevenue ELSE 0 END) AS Q2,

SUM(CASE WHEN Quarter_ = 3 THEN QuarterRevenue ELSE 0 END) AS Q3,

SUM(CASE WHEN Quarter_ = 4 THEN QuarterRevenue ELSE 0 END) AS Q4

FROM

revenue_by_quarter_and_month rqm

GROUP BY

rqm.Year_

ORDER BY

rqm.Year_;

Result Grid					
		Filter Rows:		Export:	Wrap Cell Conte
	Year_	Q1	Q2	Q3	Q4
▶	2013	0	0	84437.5	141861
	2014	147879.9	151611.09	165179.64	193718.12
	2015	315242.12	154529.22	0	0

-- 12. Retention kupaca

-- 12. Customer retention

-- Pronađi kupce koji su pravili narudžbine u 2014. godini, ali nisu u 2015. koristeći CTE i JOIN.

-- Find customers who placed orders last year but not this year using CTE and JOIN.

WITH customer_monthly_revenue_in_2015 AS (

SELECT

```

        c.companyName,
        YEAR(o.orderDate) AS Year_,
        SUM(od.quantity * od.unitPrice) AS TotalRevenue
FROM orders o
LEFT JOIN order_details od ON o.orderID = od.orderID
LEFT JOIN customers c ON o.customerID = c.customerID
GROUP BY
    c.companyName,
    YEAR(o.orderDate)
    HAVING
        Year_ = 2015
),
customer_monthly_revenue_in_2014 AS (
    SELECT
        c.companyName,
        YEAR(o.orderDate) AS Year_,
        SUM(od.quantity * od.unitPrice) AS TotalRevenue
FROM orders o
LEFT JOIN order_details od ON o.orderID = od.orderID
LEFT JOIN customers c ON o.customerID = c.customerID
GROUP BY
    c.companyName,
    YEAR(o.orderDate)
    HAVING
        Year_ = 2014
)

```

```

SELECT

```

```

*

FROM

    customer_monthly_revenue_in_2014 cmr14

LEFT JOIN

    customer_monthly_revenue_in_2015 cmr15 ON cmr14.companyName =
cmr15.companyName

WHERE

    cmr15.companyName IS NULL

ORDER BY

    cmr14.TotalRevenue DESC;

```

Result Grid


Filter Rows:

Export:



Wrap Cell Content:



	companyName	Year_	TotalRevenue	companyName	Year_	TotalRevenue
▶	Mère Paillarde	2014	26087.1	NULL	NULL	NULL
	Folies gourmandes	2014	11666.9	NULL	NULL	NULL
	Familia Arquibaldo	2014	3344.9	NULL	NULL	NULL
	Hungry Coyote Import Store	2014	2283.2	NULL	NULL	NULL
	GROSELLA-Restaurante	2014	387.5	NULL	NULL	NULL
	Woon Fat Seafood Restaurant	2014	378.0	NULL	NULL	NULL

Result 47

×

-- 13. Mesečni rast obrađenih porudžbina od strane zaposlenih

-- 13. Monthly growth of orders done by employees

-- Prikaži koliko je zaposlenih obrađivalo narudžbina po mesecu i izračunaj mesečni rast broja narudžbina zaposlenih koristeći window funkcije.

-- Show the number of orders handling by employees per month and calculate monthly orders growth using window functions.

```

WITH orders_by_employees_months AS (

```

```

    SELECT

        e.employeeName,

        YEAR(o.orderDate) AS Year_,

        MONTH(o.orderDate) AS Month_,

```

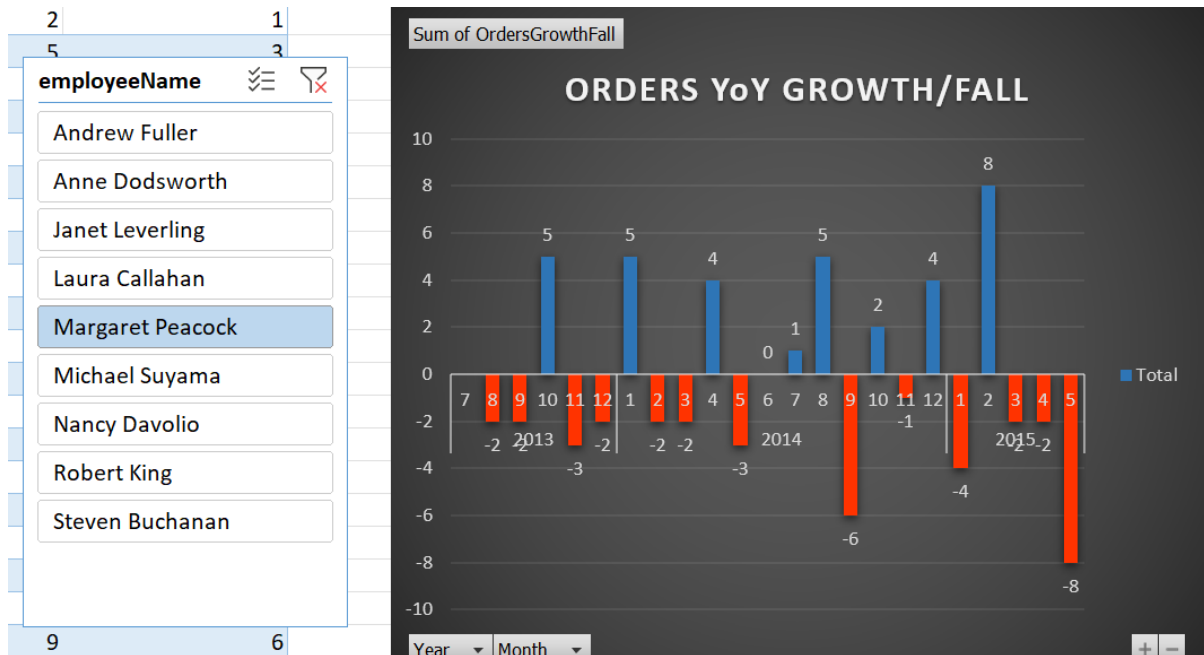
```

COUNT(o.orderID) AS TotalOrders
FROM orders o
LEFT JOIN employees e ON o.employeeID = e.employeeID
GROUP BY
    e.employeeName,
    YEAR(o.orderDate),
    MONTH(o.orderDate)
)
SELECT
    oem.employeeName,
    Year_,
    Month_,
    TotalOrders,
    (TotalOrders - LAG(TotalOrders) OVER (PARTITION BY oem.employeeName ORDER BY
    Year_,Month_)) AS OrdersGrowthFall
FROM
    orders_by_employees_months oem
ORDER BY
    oem.employeeName,
    Year_,
    Month_;

```

Result Grid					
Filter Rows:		Export:		Wrap Cell Content:	
	employeeName	Year_	Month_	TotalOrders	OrdersGrowthFall
▶	Andrew Fuller	2013	7	1	NULL
	Andrew Fuller	2013	8	2	1
	Andrew Fuller	2013	9	5	3
	Andrew Fuller	2013	10	2	-3
	Andrew Fuller	2013	11	2	0
	Andrew Fuller	2013	12	4	2

Result 48 x



-- 14. Najveći doprinos po zaposlenom

-- 14. Top contributor per employee

-- Za svaku godinu prikaži zaposlenog koji je generisao najveći prihod koristeći CTE i RANK() OVER(PARTITION BY YEAR).

-- For each year, show the employee generating the highest revenue using CTE and RANK() OVER(PARTITION BY YEAR).

WITH revenue_per_name_in_years AS (

SELECT

e.employeeName,

YEAR(o.orderDate) AS Year_,

ROUND(SUM(od.quantity * od.unitPrice),2) AS TotalRevenue

FROM orders o

LEFT JOIN order_details od ON o.orderID = od.orderID

LEFT JOIN employees e ON o.employeeID = e.employeeID

GROUP BY

e.employeeName,

```

        YEAR(o.orderDate)

    )

SELECT

    employeeName,

    Year_,

    TotalRevenue,

    RANK() OVER(PARTITION BY Year_ ORDER BY TotalRevenue DESC) AS
    RANKING_REVENUE_IN_EVERY_YEAR

FROM

    revenue_per_name_in_years

GROUP BY

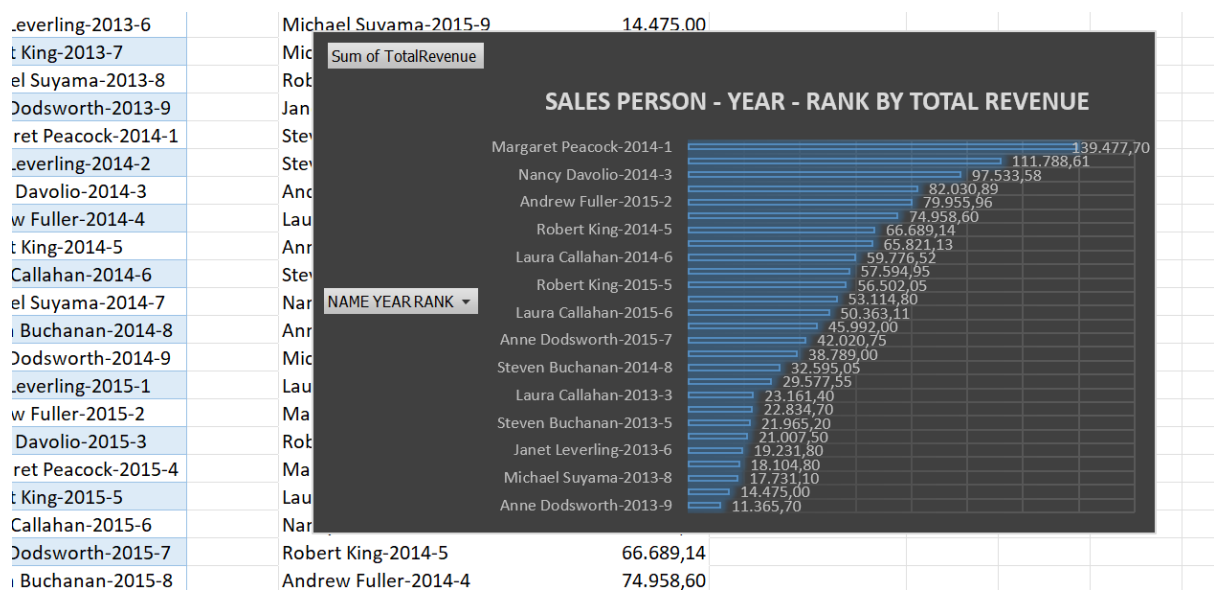
    employeeName,

    Year_,

    TotalRevenue;

```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	employeeName	Year_	TotalRevenue	RANKING_REVENUE_IN_EVERY_YEAR
▶	Margaret Peacock	2013	53114.8	1
	Nancy Davolio	2013	38789	2
	Laura Callahan	2013	23161.4	3
	Andrew Fuller	2013	22834.7	4
	Steven Buchanan	2013	21965.2	5
	Janet Leverling	2013	19231.8	6



-- 15. Analiza tezine tovara po kupcu

-- 15. Freight analysis per customer

-- Pronađi prosečanu težinu tovara po kupcu po mesecu i identifikuj kupce sa iznadprosečnim tovarom koristeći CTE.

-- Find average freight per customer per month and identify customers with above-average freight using CTE.

WITH CustomerMonthlyAverageFreight AS (

SELECT

c.companyName,

YEAR(orderDate) as Year_,

MONTH(orderDate) as Month_,

AVG(freight) as AverageFreight

FROM

orders o

LEFT JOIN

customers c ON o.customerID = c.customerID

GROUP BY

c.companyName,

YEAR(orderDate),

MONTH(orderDate)

),

MonthlyAverageFreight AS (

SELECT

YEAR(orderDate) as Year_,

MONTH(orderDate) as Month_,

AVG(freight) as AverageFreight

FROM

orders o

```

GROUP BY

YEAR(orderDate),

MONTH(orderDate)

)

SELECT

    cmaf.companyName,

    cmaf.Year_,

    cmaf.Month_,

    cmaf.AverageFreight,

    maf.AverageFreight AS AverageFreightForAll

FROM

    CustomerMonthlyAverageFreight cmaf

CROSS JOIN

    MonthlyAverageFreight maf ON cmaf.Year_ = maf.Year_ AND cmaf.Month_ =
    maf.Month_ AND cmaf.AverageFreight > maf.AverageFreight

ORDER BY

    cmaf.Year_,

    cmaf.Month_,

    cmaf.companyName;

```

Result Grid		Filter Rows:		Export:	Wrap Cell Content:
	companyName	Year_	Month_	AverageFreight	AverageFreightForAll
▶	Ernst Handel	2013	7	143.285	58.55363636363636
	Frankenversand	2013	7	208.58	58.55363636363636
	GROSELLA-Restaurante	2013	7	66.29	58.55363636363636
	Hanari Carnes	2013	7	62	58.55363636363636
	HILARION-Abastos	2013	7	81.91	58.55363636363636
	Righter Supermarket	2013	7	148.22	58.55363636363636

Result 49

Result 49 x

- companyName
- Hanari Carnes

HILARION-Abastos

Hungry Coyote Imp...

Hungry Owl All-Nigh...

Island Trading

Königlich Essen

La maison d'Asie

Lehmans Marktstand

Let's Stop N Shop

LILA-Supermercado

LINO-Delicateses

Magazzini Alimentar...

Maison Dewey

Mère Paillarde

Morgenstern Gesun...

Océano Atlántico Ltda.

Age of AverageFreight Average of AverageFreightForAll

148,69 68,16
118,40 58,55

